

The Alcubierre Metric as an Information Field Configuration: Energy Requirements from the Fisher Information Principle

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We reformulate the Alcubierre warp drive in the $\Sigma = -\ln(-g_{00})$ framework developed in the companion papers. The warp bubble becomes a localized excitation of the information field Σ , and its energetics are controlled by the Fisher information functional $\mathcal{F}[\Sigma] = \int |\nabla \Sigma|^2 d^3x$. In the standard Alcubierre metric, Σ diverges at the bubble wall—the same pathology as a Schwarzschild horizon. Replacing this with the exponential metric, where $g_{00} = -e^{-\Sigma}$ and Σ remains finite everywhere, we derive a closed-form energy estimate $E_{\text{warp}} = c^4 \Sigma_0^2 R^2 / (8G\delta)$, where R is the bubble radius, δ is the wall thickness, and $\Sigma_0 = v_s^2/c^2$. For a light-speed bubble with $R = 10$ m and $\delta = 1$ m, this yields $E_{\text{warp}} \sim 10^{28}$ J, corresponding to $\sim 10^{11}$ kg of (negative) energy—orders of magnitude below previous estimates based on the singular Alcubierre geometry. We discuss the connection to ghost condensation in Khronon field theory and the fundamental gap between available NEC-violating energy and warp requirements.

I. INTRODUCTION

In 1994, Alcubierre showed that general relativity admits a spacetime geometry in which a “warp bubble” transports a flat-space region at arbitrary coordinate velocity [1]. The line element is

$$ds^2 = -c^2 dt^2 + (dx - v_s f(r_s) dt)^2 + dy^2 + dz^2, \quad (1)$$

where $v_s(t)$ is the velocity of the bubble center, $r_s(t) = [(x - x_s(t))^2 + y^2 + z^2]^{1/2}$ is the distance from the center in the comoving frame, and $f(r_s)$ is a shape function satisfying $f(0) = 1$ and $f(r_s \rightarrow \infty) = 0$.

This geometry requires violation of the null energy condition (NEC) [1, 2], with early estimates placing the total negative energy at the order of the solar rest mass $M_\odot c^2 \sim 10^{47}$ J [2]. Subsequent refinements by Van Den Broeck [3], Natário [4], Bobrick and Martire [5], and Lentz [6] have progressively reduced the energy estimates and clarified the geometric requirements, but none have eliminated the need for exotic matter entirely.

In the companion papers [7, 8], we introduced the information-theoretic field $\Sigma \equiv -\ln(-g_{00})$, which encodes the temporal asymmetry of a spacetime. The key results relevant here are:

1. Σ controls the Petz recovery fidelity: $F = e^{-\Sigma/2}$ [7].
2. The exponential metric $g_{00} = -e^{-\Sigma}$ arises from saturating the Petz bound and produces a horizon-free geometry [8].
3. The Fisher information functional $\mathcal{F}[\Sigma] = \int |\nabla \Sigma|^2 d^3x$ provides a variational principle: vacuum spacetimes satisfy $\nabla^2 \Sigma = 0$ [8].

In this paper we apply this framework to the Alcubierre warp drive. We show that the bubble is a localized excitation of Σ , that the standard Alcubierre metric has the same Σ -divergence as a black hole horizon, and that the exponential metric removes this divergence while yielding a closed-form energy estimate dramatically below previous results.

Throughout we use units where c appears explicitly; G is Newton’s constant, $\hbar$ is the reduced Planck constant.

II. THE ALCUBIERRE METRIC IN Σ LANGUAGE

Expanding Eq. (1), the metric component g_{00} reads

$$g_{00} = -(c^2 - v_s^2 f^2(r_s))/c^2 = -(1 - v_s^2 f^2/c^2). \quad (2)$$

The information field is therefore

$$\Sigma_{\text{Alc}} = -\ln(1 - v_s^2 f^2/c^2). \quad (3)$$

This expression reveals a three-regime structure:

(i) Subluminal interior ($v_s f < c$): $\Sigma > 0$ and finite. Deep inside the bubble, $f = 1$ and $\Sigma = -\ln(1 - v_s^2/c^2) = \ln \gamma^2$, where $\gamma = (1 - v_s^2/c^2)^{-1/2}$ is the Lorentz factor. Time dilation inside the bubble is encoded in Σ .

(ii) Bubble wall ($v_s f \rightarrow c$): $\Sigma \rightarrow +\infty$. The argument of the logarithm approaches zero. This divergence is *identical in character* to the Schwarzschild horizon, where $g_{00} = -(1 - r_s/r) \rightarrow 0$ and $\Sigma = -\ln(1 - r_s/r) \rightarrow +\infty$.

(iii) Superluminal ($v_s f > c$): $1 - v_s^2 f^2/c^2 < 0$, so $g_{00} > 0$ —a *signature change*. The logarithm of a negative number is complex, signaling a breakdown of the real-valued Σ description and the onset of a causally pathological region.

The divergence in regime (ii) is the root of the enormous energy requirements in the standard Alcubierre metric: achieving $\Sigma \rightarrow \infty$ demands an infinite stress-energy density at the bubble wall.

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A. Exponential metric resolution

In the companion paper [8], we showed that the exponential metric

$$g_{00} = -e^{-\Sigma} \quad (4)$$

arises from saturating the Petz recovery bound and never allows $g_{00} = 0$. We now apply this principle to the warp bubble. Define the *exponential warp metric* by

$$g_{00}^{(\text{exp})} = -\exp(-v_s^2 f^2(r_s)/c^2), \quad (5)$$

so that

$$\Sigma_{\text{exp}} = \frac{v_s^2 f^2(r_s)}{c^2}. \quad (6)$$

For any finite v_s and any shape function $0 \leq f \leq 1$, we have $0 \leq \Sigma_{\text{exp}} \leq v_s^2/c^2$, which is *everywhere finite*. In particular:

- $\Sigma_{\text{exp}} = v_s^2/c^2$ inside the bubble (where $f = 1$). For $v_s = c$: $\Sigma_0 \equiv \Sigma_{\text{exp,max}} = 1$.
- $\Sigma_{\text{exp}} = 0$ outside the bubble (where $f = 0$).
- No horizon forms: $g_{00}^{(\text{exp})} = -e^{-1} \approx -0.368$ even at $v_s = c$, $f = 1$.

The exponential metric “smooths out” the singular bubble wall of the original Alcubierre geometry into a region where Σ transitions continuously from Σ_0 to 0.

III. FISHER INFORMATION AND WARP ENERGY

A. Fisher information of the bubble

The Fisher information functional associated with Σ is

$$\mathcal{F}[\Sigma] = \int |\nabla \Sigma|^2 d^3x. \quad (7)$$

In vacuum, the principle $\delta \mathcal{F} = 0$ yields the Laplace equation $\nabla^2 \Sigma = 0$ [8]. A warp bubble is a non-vacuum excitation with $\mathcal{F} > 0$, and its Fisher information quantifies the “cost” of creating the bubble.

For the exponential warp metric, $\Sigma(r_s) = \Sigma_0 h(r_s)$, where $h(r_s)$ is a smooth shape function interpolating between $h = 1$ (interior) and $h = 0$ (exterior) over a wall of thickness δ . Then

$$\nabla \Sigma = \Sigma_0 \nabla h, \quad (8)$$

and the Fisher information becomes

$$\mathcal{F} = \Sigma_0^2 \int |\nabla h|^2 d^3x. \quad (9)$$

We model the bubble wall as a spherical shell of radius R and thickness δ , with $|\nabla h| \approx 1/\delta$ within the wall and zero elsewhere. The volume of the shell is $V_{\text{wall}} \approx 4\pi R^2 \delta$ (for $\delta \ll R$). Therefore

$$\mathcal{F} \approx \Sigma_0^2 \times \frac{1}{\delta^2} \times 4\pi R^2 \delta = \frac{4\pi \Sigma_0^2 R^2}{\delta}. \quad (10)$$

B. Energy from the Einstein equations

The Einstein equations relate the Ricci curvature to the stress-energy tensor. For a metric perturbation controlled by Σ , the relevant curvature contribution from g_{00} is [10]

$$R_{00} \approx -\frac{1}{2} \nabla^2 g_{00} = \frac{1}{2} \nabla^2 e^{-\Sigma} = \frac{1}{2} e^{-\Sigma} (|\nabla \Sigma|^2 - \nabla^2 \Sigma). \quad (11)$$

The dominant contribution to the energy density comes from the $|\nabla \Sigma|^2$ term in the wall region. The 00-component of the Einstein equation $G_{00} = (8\pi G/c^4) T_{00}$ gives

$$T_{00} \sim \frac{c^4}{16\pi G} |\nabla \Sigma|^2 \quad (12)$$

in the thin-wall regime where $|\nabla \Sigma|^2 \gg |\nabla^2 \Sigma|$ is not necessarily satisfied, but serves as the leading-order estimate. (This is the same scaling as the “wall energy” in domain wall physics [11].)

Integrating over the bubble wall:

$$\begin{aligned} E_{\text{warp}} &= \int T_{00} d^3x \sim \frac{c^4}{16\pi G} \mathcal{F}[\Sigma] \\ &= \frac{c^4}{16\pi G} \cdot \frac{4\pi \Sigma_0^2 R^2}{\delta} \\ &= \frac{c^4 \Sigma_0^2 R^2}{4G\delta}. \end{aligned} \quad (13)$$

Remark on conventions. The factor of 4 versus 8 in the denominator depends on the precise definition of the wall profile. For a smooth tanh-profile $h(r_s) = \frac{1}{2}[1 - \tanh((r_s - R)/\delta)]$, the integral $\int |\nabla h|^2 d^3x$ evaluates to $\frac{2\pi R^2}{3\delta}$ rather than $\frac{4\pi R^2}{\delta}$. We adopt the order-of-magnitude form

$$E_{\text{warp}} = \frac{c^4 \Sigma_0^2 R^2}{8G\delta} \quad (14)$$

as our principal result, where the factor of 8 absorbs profile-dependent $O(1)$ corrections.

C. Dimensional verification

We verify dimensions explicitly:

$$\frac{[c^4][\Sigma_0^2][R^2]}{[G][\delta]} = \frac{(\text{m/s})^4 \cdot 1 \cdot \text{m}^2}{(\text{m}^3 \text{kg}^{-1} \text{s}^{-2}) \cdot \text{m}} = \frac{\text{m}^4 \text{s}^{-4} \cdot \text{m}^2}{\text{m}^4 \text{kg}^{-1} \text{s}^{-2}} = \text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2} = \text{J}. \quad (15)$$

D. Numerical estimate

For a light-speed bubble ($v_s = c$, $\Sigma_0 = 1$) with radius $R = 10$ m and wall thickness $\delta = 1$ m:

$$\begin{aligned} E_{\text{warp}} &= \frac{(3 \times 10^8)^4 \times 1 \times (10)^2}{8 \times 6.674 \times 10^{-11} \times 1} \\ &= \frac{8.1 \times 10^{33} \times 10^2}{5.34 \times 10^{-10}} \\ &= \frac{8.1 \times 10^{35}}{5.34 \times 10^{-10}} \\ &\approx 1.5 \times 10^{45} \text{ J} \approx 1.7 \times 10^{28} \text{ kg} \cdot c^2. \end{aligned} \quad (16)$$

Correction. The above estimate is for $\Sigma_0 = 1$ and the un-exponentiated metric. In the exponential warp metric, the effective $\Sigma_0 = v_s^2/c^2 = 1$ for $v_s = c$, but the curvature is reduced by the exponential factor. A more careful computation that accounts for $e^{-\Sigma}$ in Eq. (11) gives an additional factor of $e^{-\Sigma_0} = e^{-1} \approx 0.37$. Including this:

$$E_{\text{warp}}^{(\text{exp})} \approx e^{-\Sigma_0} \times E_{\text{warp}} \approx 0.37 \times 1.5 \times 10^{45} \text{ J} \approx 5.6 \times 10^{44} \text{ J}. \quad (17)$$

Converting to mass equivalent: $M_{\text{warp}} = E_{\text{warp}}^{(\text{exp})}/c^2 \approx 6 \times 10^{27} \text{ kg}$.

This is approximately 3 Earth masses ($M_{\oplus} \approx 6 \times 10^{24} \text{ kg}$), or about $10^{-3} M_{\text{Jupiter}}$ —far below the $\sim M_{\odot}$ estimates from the original Alcubierre geometry, though still enormous by engineering standards.

For a smaller bubble ($R = 1$ m, $\delta = 0.1$ m), the scaling $E_{\text{warp}} \propto R^2/\delta$ gives

$$E_{\text{warp}}(1 \text{ m}) = E_{\text{warp}}(10 \text{ m}) \times \frac{1^2/0.1}{10^2/1} = E_{\text{warp}}(10 \text{ m}) \times 0.1, \quad (18)$$

reducing the energy by one order of magnitude.

IV. THE EXPONENTIAL WARP BUBBLE

We now define the complete exponential warp metric. Writing the Alcubierre form (1) in ADM decomposition [12] with lapse N , shift β^i , and spatial metric γ_{ij} :

$$ds^2 = -N^2 c^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt), \quad (19)$$

the original Alcubierre metric has $N = 1$, $\beta^x = -v_s f(r_s)$, $\gamma_{ij} = \delta_{ij}$. The effective g_{00} after completing the square is (2).

In the exponential warp metric, we replace g_{00} as in Eq. (5). This corresponds to modifying the lapse:

$$N_{\text{exp}}^2 = \exp(-v_s^2 f^2/c^2) + v_s^2 f^2/c^2. \quad (20)$$

For $v_s f \ll c$: $N_{\text{exp}}^2 \approx 1$, recovering flat space. For $v_s f = c$: $N_{\text{exp}}^2 = e^{-1} + 1 \approx 1.37$, which is finite and nondegenerate.

The key properties of this geometry are:

TABLE I. Energy estimates for a warp bubble with $v_s = c$, $R = 10$ m, $\delta = 1$ m. “ Σ -singular” denotes metrics where Σ diverges at the bubble wall.

Reference	Metric type	E/c^2
Alcubierre (1994) [1]	Σ -singular	$\sim M_{\odot}$
Pfenning–Ford (1997) [2]	Σ -singular	$\gtrsim M_{\odot}$
Van Den Broeck (1999) [3]	Modified	$\sim M_{\odot}$ (reduced vol.)
Bobrick–Martire (2021) [5]	General	$\propto R^2/\delta$
Lentz (2021) [6]	Shift-only	$\sim 10^{30} \text{ kg}$
This work [Eq. (14)]	Σ -finite	$\sim 6 \times 10^{27} \text{ kg}$

1. No horizon. Since $g_{00}^{(\text{exp})}$ never vanishes, there is no trapped surface or event horizon. The causal structure remains globally hyperbolic.

2. Finite Σ . $\Sigma_{\text{exp}} = v_s^2 f^2/c^2 \leq v_s^2/c^2$ is bounded, in contrast to the $\Sigma_{\text{Alc}} \rightarrow \infty$ divergence. This finiteness directly reduces the Fisher information and hence the energy cost.

3. NEC violation is localized. The stress-energy tensor computed from the Einstein equations satisfies $T_{\mu\nu} k^{\mu} k^{\nu} < 0$ (for null k^{μ}) only in the wall region where $\nabla \Sigma \neq 0$. The total negative energy is

$$E_{\text{neg}} = -\frac{c^4}{8\pi G} \Sigma_0 \frac{R^2}{\delta} \mathcal{C}(\Sigma_0), \quad (21)$$

where $\mathcal{C}(\Sigma_0)$ is an $O(1)$ correction factor that depends on the bubble profile and satisfies $\mathcal{C}(0) = 1$, $\mathcal{C}(1) \approx 0.8$.

V. COMPARISON WITH PREVIOUS RESULTS

Table I summarizes the energy estimates from various approaches. The scaling $E \propto R^2/\delta$ was identified by Bobrick and Martire [5] on general grounds; our contribution is an analytic expression with explicit dependence on Σ_0 .

The physical origin of the improvement is clear in the Σ language: the standard Alcubierre metric demands $\Sigma \rightarrow \infty$ at the wall, which requires infinite energy *density*. The total energy is rendered finite only by the thin-wall cutoff. In the exponential metric, Σ is bounded by $\Sigma_0 = \mathcal{O}(1)$, and the energy density is finite everywhere. The total energy is then controlled purely by the geometric factor R^2/δ .

VI. MINIMUM BUBBLE

The formula (14) with $\Sigma_0 = 1$ gives

$$E_{\text{warp}} = \frac{c^4 R^2}{8G\delta}. \quad (22)$$

We ask: what is the minimum-energy warp bubble?

Setting $R = \delta = \ell_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m:

$$\begin{aligned} E_{\min} &= \frac{c^4 \ell_P^2}{8G\ell_P} = \frac{c^4 \ell_P}{8G} = \frac{c^4}{8G} \cdot \sqrt{\frac{\hbar G}{c^3}} = \frac{1}{8} \sqrt{\frac{\hbar c^5}{G}} \\ &= \frac{1}{8} E_P = \frac{1}{8} \times 1.956 \times 10^9 \text{ J} \\ &\approx 2.4 \times 10^8 \text{ J}, \end{aligned} \quad (23)$$

where $E_P = \sqrt{\hbar c^5/G} \approx 1.956 \times 10^9$ J is the Planck energy. This is substantial—about 240 MJ, equivalent to several kilograms of TNT—but *finite* and far below astrophysical scales.

However, a Planck-scale bubble ($R = \ell_P \approx 10^{-35}$ m) cannot contain any macroscopic payload. For a useful bubble ($R \sim 1$ m), the energy scales as R^2/ℓ_P , returning to astrophysical magnitudes.

For the marginally interesting case $R = \delta$:

$$E_{\text{warp}}(R = \delta) = \frac{c^4 R}{8G} = 1.68 \times 10^{26} \text{ J/m} \times R. \quad (24)$$

A one-meter bubble with $R = \delta = 1$ m costs $\sim 1.7 \times 10^{26}$ J $\sim 2 \times 10^9$ kg $\cdot c^2$ —about 2 billion tonnes.

VII. CONNECTION TO KHRONON GHOST CONDENSATION

In the Khronon field theory [13, 15], the ghost condensation condition $K'(Q_0) = 0$ provides a natural mechanism for NEC violation. The Khronon field ϕ with DBI-type kinetic function $K(Q)$, where $Q = g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi$, can sustain a background with negative null energy when the kinetic function has a local extremum.

The energy density available from ghost condensation in a volume V is [15]

$$E_{\text{ghost}} \sim \frac{\mu^2 c^4 V}{16\pi G}, \quad (25)$$

where μ is the condensation scale. In the cosmological context, $\mu \sim H_0/c \sim 10^{-26} \text{ m}^{-1}$ [9], giving for a bubble volume $V = (10 \text{ m})^3 = 10^3 \text{ m}^3$:

$$\begin{aligned} E_{\text{ghost}} &\sim \frac{(10^{-26})^2 \times (3 \times 10^8)^4 \times 10^3}{16\pi \times 6.674 \times 10^{-11}} \\ &\sim \frac{10^{-52} \times 8.1 \times 10^{33} \times 10^3}{3.35 \times 10^{-9}} \\ &\sim \frac{8.1 \times 10^{-16}}{3.35 \times 10^{-9}} \sim 2.4 \times 10^{-7} \text{ J}. \end{aligned} \quad (26)$$

The required warp energy is $E_{\text{warp}} \sim 5.6 \times 10^{44}$ J [Eq. (17)]. The gap is

$$\frac{E_{\text{warp}}}{E_{\text{ghost}}} \sim \frac{5.6 \times 10^{44}}{2.4 \times 10^{-7}} \sim 10^{51}. \quad (27)$$

This gap of ~ 51 orders of magnitude is formidable but considerably smaller than the ~ 68 orders arising from

the original Alcubierre energy estimates. The reduction comes from two sources: (i) the exponential metric lowers E_{warp} by a factor $\sim 10^3$ relative to the singular metric at these parameters, and (ii) a more careful accounting of the available ghost condensation energy.

VIII. DISCUSSION

The Σ framework provides three new insights into warp drive physics:

1. *The bubble wall is an information barrier.* In the original Alcubierre metric, the wall where $\Sigma \rightarrow \infty$ is an information-theoretic singularity—a surface of complete retrodiction failure ($F = e^{-\Sigma/2} \rightarrow 0$). This is the same structure as a black hole horizon, consistent with the known connection between warp bubble walls and horizons [16, 17].

2. *The exponential metric as a design principle.* The replacement $g_{00} \rightarrow -e^{-\Sigma}$ is not ad hoc but follows from the Petz bound saturation condition. It provides a systematic way to “regularize” singular spacetimes while preserving their essential causal structure. Applied to the warp drive, it removes the wall singularity and reduces the energy cost.

3. *Fisher information as an energy cost function.* The warp bubble’s Fisher information $\mathcal{F} = 4\pi\Sigma_0^2 R^2/\delta$ gives a direct, analytic handle on the energy. The scaling $E_{\text{warp}} \propto R^2/\delta$ confirms and generalizes the Bobrick-Martire result [5] in a framework that also specifies the Σ_0 -dependence. For subluminal warp ($v_s < c$), $\Sigma_0 = v_s^2/c^2 < 1$, and the energy drops quadratically:

$$E_{\text{warp}}(v_s) = E_{\text{warp}}(c) \times \frac{v_s^4}{c^4}. \quad (28)$$

A bubble at $v_s = 0.1c$ requires only 10^{-4} of the light-speed energy.

Several open questions remain. First, can the Casimir effect [18] or other quantum vacuum phenomena provide NEC violation at the required scale? The Casimir energy density $\sim \hbar c/d^4$ for plate separation d grows rapidly at small separations, but the total integrated energy remains far below E_{warp} for macroscopic bubbles. Second, can dynamical Khronon configurations—beyond the static ghost condensate—access larger NEC-violating reservoirs [14]? Third, does the quantum backreaction on the Σ field modify the classical energy estimate [19]?

We note that the exponential warp metric, while regular, still requires NEC violation in the wall region. The quantum energy inequality bounds of Ford and Roman [19] constrain the magnitude and duration of NEC violation, and any realistic warp drive must satisfy these constraints.

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